

AI EXPRESS HACKATHON: TECHNICAL SUMMARY SHEET

Track: Hybrid Autonomous Mars Rover Agent (Logic KB + A* Search Evasion)

Course Code: AI-401	Group ID: Team Alpha
Members: Student 1, Student 2, Student 3	Track: Hybrid AI Agent (Logic KB + A* Search)
GitHub Repo: https://github.com/hackathon/hybrid-mars-rover	Environment: 10x10 Fog-of-War Grid (Pygame)

1. PEAS Framework Matrix

Parameter	Description
Performance	Reach Goal G safely, 0 steps into hazards/radiation, minimize path cost, optimize algorithm handoffs.
Environment	10x10 2D grid, partially observable (fog of war), deterministic, multi-cell storm hazards, discrete.
Actuators	Orthogonal movement execution: Move Up, Move Down, Move Left, Move Right.
Sensors	Local Breeze sensor B(x,y) [Hazard/Storm], Local Glow sensor G(x,y) [Radiation], Position sensor (x,y).

2. Hybrid Algorithmic Formulation (Units 1 - 4)

Primary Engine (Propositional Logic KB - Units 3&4): Proves safe cells $KB \models S(x,y)$ using Unit Resolution and Model Entailment over atomic propositions H, R, B, G, S, V.

Evasion Engine (A* Search - Units 1&2): When a Multi-Cell Storm or Barrier is detected, the agent dynamically switches algorithms to A* Graph Search using Manhattan distance heuristic $h(n) = |dx| + |dy|$ and evaluation function $f(n) = g(n) + h(n)$.

3. Theoretical & Observed Complexity Analysis

Metric	Theoretical (Big-O)	Observed Execution
Logic Time Complexity	$O(N^2 * k)$ polynomial unit resolution per step	~0.01 - 0.02 seconds per step inference
A* Time Complexity	$O(N^2 \log N)$ priority queue graph search	12-18 nodes expanded per A* evasion run
Space Complexity	$O(N^2)$ for KB literal stores & A* open/closed sets	Max 100 states stored in memory
Algorithm Handoffs	Dynamic mode switch on storm barrier detection	Seamless handoff KB --> A* Search